

## §7 Designing with Logic

### 7.1 Device Family Overview

#### ALVC Family

- The highest performance 3.3-V bus-interface in 0.6- $\mu$  CMOS technology
- Typical propagation delays of less than 3 ns
- Current drive of 24 mA and static power consumption of 40 mA for bus-interface functions
- Bus-hold cells on inputs (no need for external pullup resistors for floating inputs)

#### ABT Family

- The ABT family is manufactured using the latest 0.8- $\mu$  BiCMOS process
- It provides high drive up to 64 mA and propagation delays below the 5-ns range, while maintaining very low power consumption.
- To reduce transmission-line effects, the ABT family has series-damping resistor options.
- There are special ABT parts that provide extremely high-current drive (180 mA).

#### GTL Family

- Reduced-voltage switching standard that provides high-speed, point-to-point communications
- Low power dissipation
- Feature innovative circuitry, such as bus hold on the TTL inputs, to eliminate the need for external pullup resistors for floating inputs
- Output edge-rate control (OEXE) to reduce electromagnetic interference (EMI)

#### LV Family

- 3-V power supply use with the same 5-V performance characteristics of HCMOS logic
- 2- $\mu$  CMOS process that provides up to 8 mA of drive
- Propagation delays of 18 ns maximum
- Static power consumption of only 20 mA

#### LVC Family

- 3-V power supply use, with about the same performance as the 5-V 74F family
- 0.8- $\mu$  CMOS process technology, 24-mA current drive
- 6.5-ns maximum propagation delays
- 5-V tolerant inputs and outputs

#### LVT Family

- 3-V LVT family uses the latest 0.8- $\mu$  BiCMOS process technology
- Up to 64 mA of drive and 4-ns propagation delays
- Consumes less than 100 mA of standby power

- Bus-hold feature eliminates external pullup resistors and I/Os that can handle up to 7 V, which allows them to act as 5-V/3-V translators.

#### LVTZ Family

- Offers all of the features found in standard LVT family
- Incorporates circuitry to protect the devices in live-insertion applications
- The device goes to the high-impedance state during power up and power down

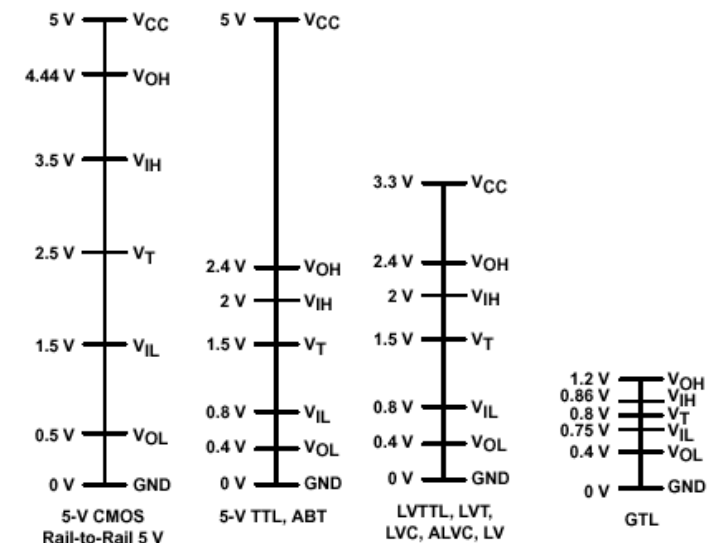


Figure 7.1 Summary of switching threshold voltages for different logic families

### 7.2 CMOS and BiCMOS Input Characteristics

- Both advanced CMOS (ALVC, LVC, and LV) and BiCMOS (LVT, GTL) families have a CMOS input structure.
- The input is an inverter consisting of a p-channel to V<sub>CC</sub> and an n-channel to GND, as shown in Figures 7.2.
- When a low level is applied to the input, the p-channel transistor is ON and the n-channel is OFF, resulting in the current flowing from V<sub>CC</sub> and pulling the node to a high state.
- When a high level is applied, the n-channel transistor is ON and the p-channel is OFF and the current flows to GND, pulling the node low.
- In both cases, no current flows from V<sub>CC</sub> to GND.

- However, when switching from one state to another, the input crosses the threshold region, causing the n-channel and the p-channel to be turned on simultaneously, generating a current path between  $V_{CC}$  and GND.
- This current surge can be damaging, depending on the length of time that the input is in the threshold region (0.8 V to 2 V). The supply current ( $I_{CC}$ ) can rise up to several milliamperes (mA) per input, peaking at approximately 1.5-V. (see Figure 7.3).

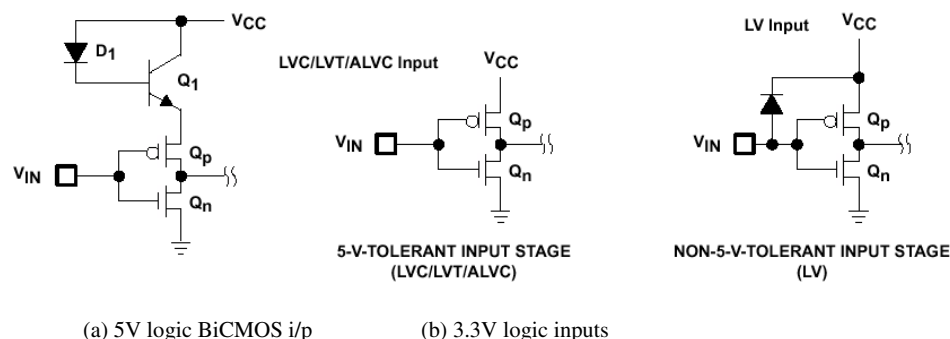


Figure 7.2 Logic input equivalent circuit

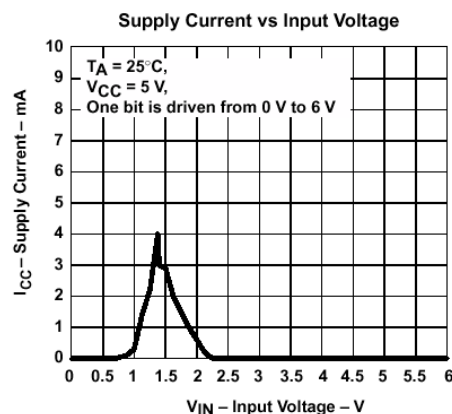


Figure 7.3 Typical transient supply current

### 7.3 BiCMOS Output Characteristics

- Figure 7.4 shows a simplified LVT output and illustrates the mixed-mode capability designed into the output stage.
- This combination of a high-drive TTL stage, along with the rail-to-rail CMOS switching, gives the LVT series exceptional application flexibility.
- These parts provide the dc drive needed for existing 5-V backplanes. Thus, using LVT is a simple way to reduce system power via the migration to 3.3-V operation.
- Not only can LVT devices operate as 3-V- to 5-V-level translators by supporting 5-V input or I/O voltages ( $V_{CC} = 2.7$  V to 3.6 V), but also the inputs can withstand 5.5V, even when  $V_{CC} = 0$  V.
- This allows for the devices to be used under partial system power-down and live-insertion applications.

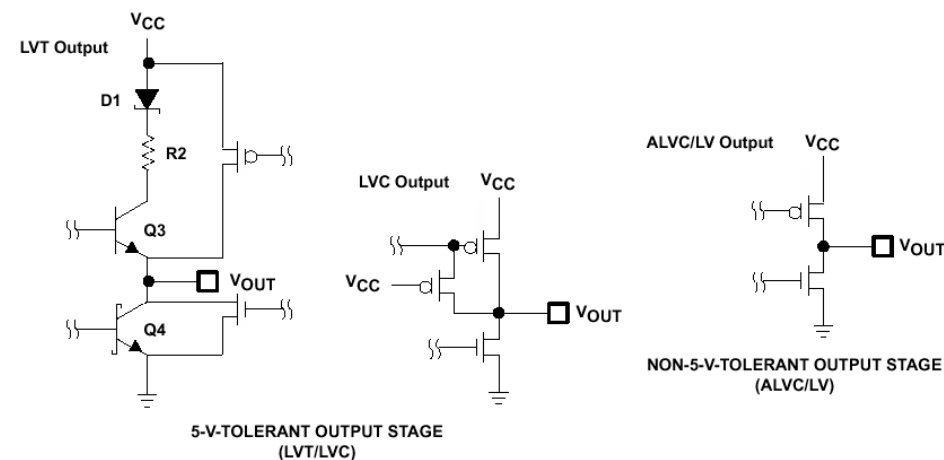


Figure 7.4 BiCMOS/CMOS outputs

### 7.4 CMOS Output Characteristics (ALVC, LVC, and LV)

Figure 7.4 also shows a simplified LV, LVC, and ALVC output stage.

- LV and ALVC are pure 3.3-V families. They cannot be used to translate between 5-V and 3.3-V environments.
- ALVC is used primarily for high-speed memory and point-to-point applications with medium drive capability ( $\pm 24$  mA).
- LV is designed for low-speed, low-drive ( $\pm 8$ –6 mA) applications. It is similar to HC and HCT.
- LVC is used for on-board and memory applications that require medium performance and medium drive logic, as well as translation between 5-V and 3.3-V signals.
- Not only can LVC devices operate as 3-V- to 5-V-level translators by supporting 5.5-V input or I/O voltages ( $V_{CC} = 3$  V to 3.6 V), but the inputs can withstand 5.5 V, even when  $V_{CC} = 0$  V.
- This permits the devices to be used under partial system power-down and live-insertion applications.

### 7.5 Power Consumption

- Several factors influence the power consumption of a device: frequency of operation, number of outputs switching, load capacitance, number of TTL-level inputs, junction temperature, ambient temperature, and thermal resistance of the device.
- The maximum operating frequency is limited by the thermal characteristics of the package.
- “absolute maximum ratings” are usually calculated using a junction temperature of 150°C and a board trace length of 750 mils (no airflow).
- Traces, power planes, connectors, and cooling fans play an important role in improving the heat dissipation.
- Figures 7.5 shows the maximum frequency at which a family can operate and still meet the  $V_{OH}$  and  $V_{OL}$  specifications. No frequency beyond the maximum number is acceptable.
- Note that all registered devices were tested based on the clock frequency, and the nonregistered devices were tested based on the input frequency.

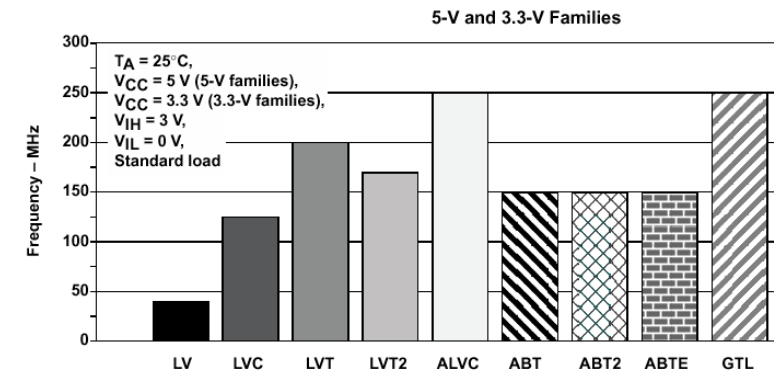


Figure 7.5 Maximum frequency of operation for different logic families (TI)

### 7.6 The importance of 3.3v families

Figure 7.6 shows the expected growth of low voltage memory devices in the near future:

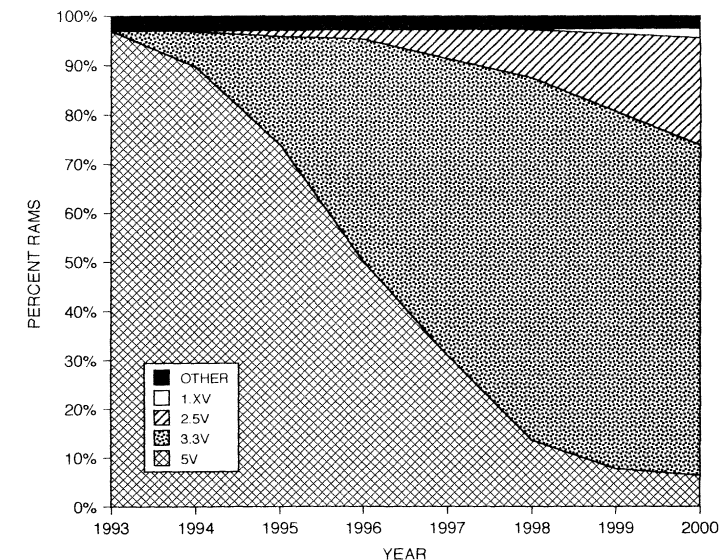


Figure 7.6 Market share of RAM devices using different voltages

Conclusion: use 3.3V or lower!

7.7 Package Power Dissipation

- Thermal awareness became an industry concern when surface-mount technology (SMT) packages began replacing through-hole (DIP) packages in PCB designs.
- Circuits operating at the same power enclosed in a smaller package meant higher power density.
- To add to the issue, systems required increased throughput, which resulted in higher frequencies, increasing the power density even further.
- Figure 7.7 explains part of the reason for increased attention to thermal issues. As a baseline for comparison, the 24-pin small-outline integrated circuit (SOIC) is shown, along with several fine-pitch packages, including the 24- and 48-pin SSOP, 24- and 48-pin TSSOP, and 100-pin thin quad flat pack (TQFP).
- The 24-pin TSSOP (8, 9, and 10 bits) allows for the same circuit functionality of the 24-pin SOIC to be packaged in less than a third of the area, while the 48-pin TSSOP (16, 18, and 20 bits) occupies less area and has twice the functionality of the 24-pin SOIC.
- This same phenomenon is expanded even further with the 100-pin TQFP (32 and 36 bits), which is the functional equivalent of four 24-pin or two 48-pin devices, with additional board savings over that of the SSOP packages.
- As the trend in packaging technology moves toward smaller packages, attention must be focused on the thermal issues that are created.

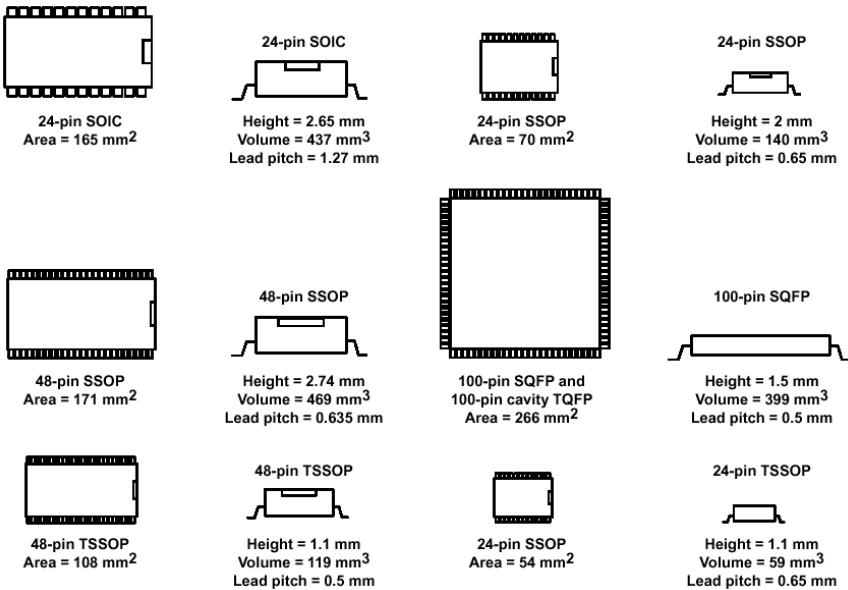


Figure 7.7 Advanced Packaging

- A better understanding of the factors that contribute to junction temperature ( $T_J$ ) provides a system designer with more flexibility when attempting to solve thermal issues.
- Device junction temperature is determined by the equation:

$$T_J = T_A + (\Theta_{JA} \times P_T)$$

where:  
 $T_J$  = Junction (die) temperature (°C)  
 $T_A$  = Ambient temperature (°C)  
 $\Theta_{JA}$  = Thermal resistance of the package from the junction to the ambient (°C/W)  
 $P_T$  = Total power of the device (W)

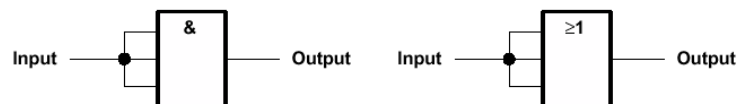
- Junction temperature is altered by lower chip power consumption, longer trace length, heatsinks, forced air flow, package mold compound, lead-frame size and material, surface area, and die size.
- Some of these are mechanically inherent in a particular package, while others are controlled by the designer and are application specific.
- Understanding which variables can be influenced by practicing good thermal-design techniques requires a more detailed investigation of power considerations as well as thermal-resistance measurements.
- The package power dissipation is calculated using a junction temperature ( $T_J$ ) of 150°C and an ambient temperature ( $T_A$ ) of 55°C.
- $\Theta_{JA}$  is calculated using a board trace length of 750 mils and no airflow.
- Figure 7.8 provides the different  $\Theta_{JA}$  for different packages.

| NO. OF PINS |               | 14  | 16  | 20  | 24  | 48  | 52 | 56  | 64 | 80 | 100  |
|-------------|---------------|-----|-----|-----|-----|-----|----|-----|----|----|------|
| SOIC        | Package       | D   | D   | DW  | DW  | —   | —  | —   | —  | —  | —    |
|             | $\Theta_{JA}$ | 76  | 73  | 59  | 56  | —   | —  | —   | —  | —  | —    |
| SSOP        | Package       | DB  | DB  | DB  | DB  | DL  | —  | DL  | —  | —  | —    |
|             | $\Theta_{JA}$ | 185 | 175 | 164 | 152 | 80  | —  | 68  | —  | —  | —    |
| TSSOP       | Package       | PW  | PW  | PW  | PW  | DGG | —  | DGG | —  | —  | —    |
|             | $\Theta_{JA}$ | 195 | 187 | 143 | 140 | 115 | —  | 92  | —  | —  | —    |
| QFP         | Package       | —   | —   | —   | —   | —   | RC | —   | —  | PH | —    |
|             | $\Theta_{JA}$ | —   | —   | —   | —   | —   | 69 | —   | —  | 84 | —    |
| TQFP        | Package       | —   | —   | —   | —   | —   | —  | —   | PM | PN | PZ   |
|             | $\Theta_{JA}$ | —   | —   | —   | —   | —   | —  | —   | 96 | 89 | 79   |
| TQFP-HP     | Package       | —   | —   | —   | —   | —   | —  | —   | —  | —  | PCA  |
|             | $\Theta_{JA}$ | —   | —   | —   | —   | —   | —  | —   | —  | —  | 52.4 |

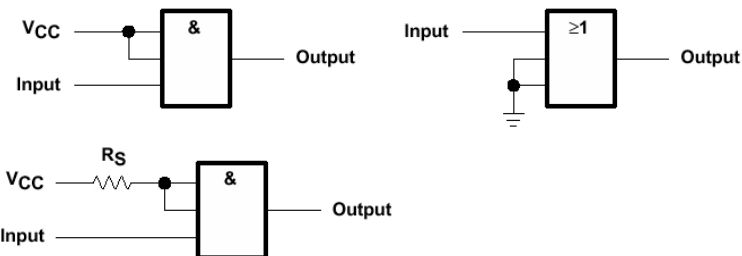
Figure 7.8  $\Theta_{JA}$  for different packages

### 7.8 Used Inputs

- All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating.
- The logic level that should be applied to any particular unused input depends on the function of the device.
- One solution is to connect unused inputs to an input of the same gate that is in use. Connecting the inputs together increases the capacitive load on the driver stage and, with bipolar circuits, also increases the dc current drain.



- A better solution is to apply a fixed logic level to the unused inputs. If a low level is required, the input should be directly connected to GND; if a high level is required, it should be connected to the positive supply voltage  $V_{CC}$  directly or via a series resistor.



### 7.9 Slow Input Edge Rate

- With increased speed, logic devices have become more sensitive to slow input edge rates.
- A slow input edge rate, coupled with the noise generated on the power rails when the output switches, can cause excessive output errors or oscillations.
- Similar situations can occur if an unused input is left floating or is not actively held at a valid logic level.
- These functional problems are due to voltage transients induced on the device's power system as the output load current flows through the parasitic lead inductances during switching (see Figure 7.9).
- Because the device's internal power-supply nodes are used as voltage references throughout the integrated circuit, inductive voltage spikes,  $V_{GND}$ , affect the way signals appear to the internal gate structures. For example, as the voltage at the

device's ground node rises, the input signal,  $V_I'$ , appears to decrease in magnitude. This undesirable phenomenon can then erroneously change the output if a threshold violation occurs.

- In the case of a slowly rising input edge, if the change in voltage at GND is large enough, the apparent signal,  $V_I'$ , at the device appears to be driven back through the threshold and the output starts to switch in the opposite direction.
- If worst-case conditions prevail (simultaneously switching all of the outputs with large transient load currents), the slow input edge is repeatedly driven back through the threshold, causing the output to oscillate.

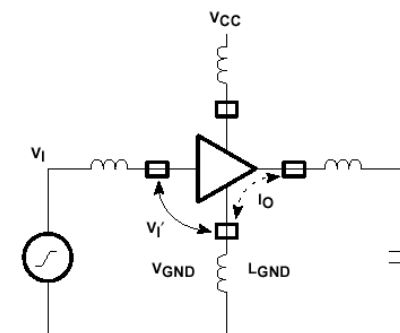


Figure 7.9 Input/output model with packaging effect

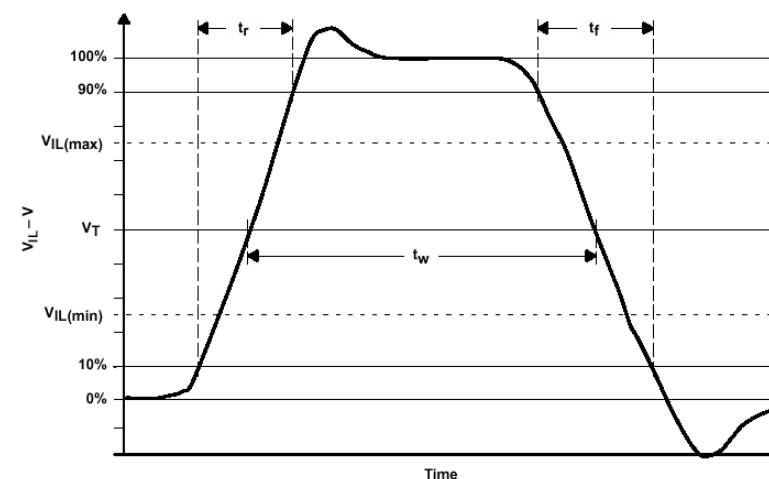


Figure 7.10 Definition of Rise and Fall times

- Therefore, correct operation of the circuit can be ensured only if the rise and fall times of the signal at the input do not exceed certain values. (See Figure 7.10)

- The signal amplitude is specified as the difference between the two stable signal levels for high ( $V_H$ ) and low ( $V_L$ ); overshoot and undershoot of the signal are not taken into account. The difference  $V_H - V_L$  is taken as 100% of the amplitude.
- The rise time of the signal is defined as the time taken to rise from 10% to 90% of the full amplitude; similarly, the fall time is the time taken to fall from 90% to 10% of the amplitude.
- The pulse width ( $t_w$ ) of a signal is measured at 50% of the amplitude. However, these definitions must be used for digital circuits with certain qualifications because, in most cases, the switching threshold ( $V_T$ ) of the input is not 50% of the amplitude. So, the level needed by the circuit must be considered and, from this, the required signal waveform derived.
- Figure 7.11 shows the necessary minimum transition rise/fall rates for various logic families.

| SERIES  | $V_{CC}$<br>(V) | $V_{IL(max)}$<br>(V) | $V_{IH(min)}$<br>(V) | $V_T$<br>(V)  | $dt/dv$<br>(ns/V) |
|---------|-----------------|----------------------|----------------------|---------------|-------------------|
| SN74    | 4.75–5.25       | 0.8                  | 2                    | 1.4           | 15                |
| SN74LS  | 4.75–5.25       | 0.8                  | 2                    | 1.4           | 15                |
| SN74S   | 4.75–5.25       | 0.8                  | 2                    | 1.4           | 15                |
| SN74ALS | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 15                |
| SN74AS  | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 15                |
| SN74F   | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 15                |
| SN74HC  | 2               | 0.3                  | 1.5                  | 1.4           | 625               |
|         | 4.6             | 0.9                  | 3.15                 | 2.25          | 110               |
|         | 6               | 1.2                  | 4.2                  | 3             | 80                |
| SN74HCT | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 125               |
| 74AC    | 3               | 0.9                  | 2.1                  | 1.5           | 10                |
|         | 4.5             | 1.35                 | 3.15                 | 2.25          | 10                |
|         | 5.5             | 1.65                 | 3.85                 | 2.75          | 10                |
| 74ACT   | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 10                |
| SN74BCT | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 10                |
| SN74ABT | 4.5–5.5         | 0.8                  | 2                    | 1.4           | 5/10              |
| SN74LV  | 2.7–3.6         | 0.8                  | 2                    | $\approx 1.5$ | 100               |
| SN74LVC | 2.7–3.6         | 0.8                  | 2                    | $\approx 1.5$ | 5/10              |
| SN74LVT | 3.0–3.6         | 0.8                  | 2                    | 1.4           | 10                |

Figure 7.11 Minimum rise/fall rate for different logic families

## 7.10 Floating Inputs

- As long as the driver is active in a transmission path or bus, the receiver's input is always in a valid state. No input specification is violated as long as the rise and fall times are within the data-sheet limits.
- However, when the driver is in a high-impedance state, the receiver input is no longer at a defined level and tends to float.
- This situation can worsen when several transceivers share the same bus.

- Figure 7.12 is an example of a typical bus system. When all transceivers are inactive, the bus-line levels are undefined.
- When a voltage that is determined by the leakage currents of each component on the bus is reached, the condition is known as a *floating state*.
- The result is a considerable increase in power consumption and a risk of damaging all components on the bus.
- Holding the inputs or I/O pins at a valid logic level when they are not being used or when the part driving them is in the high-impedance state is recommended.

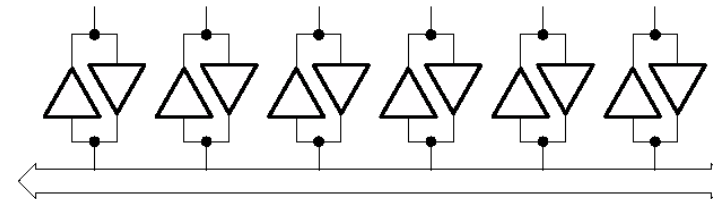
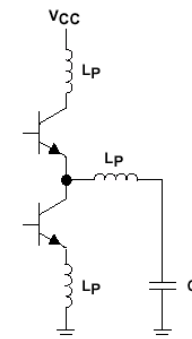


Figure 7.12 Bus system

## 7.11 Propagation Delay Times With Several Outputs Switching Simultaneously

- The propagation delay times of circuits given in data sheets apply when only one output switches at a time. The reason for this is that the production equipment used to test circuits can test only one transmission channel at a time.
- If several outputs switch simultaneously, the propagation delay times given in data sheets can be used only with reservations. The reason for this is that the package inductances ( $L_P$ ) of the supply-voltage lines, as well the output lines (see figure below), have a significant influence on the circuits and, thus, on the delay times.
- These inductances have the effect that the current in the power supply lines, and consequently in the output of the device, has a limited rate of rise.
- For this reason, when several outputs switch simultaneously, only a limited output current is available.
- Figure 7.13 shows the influence on the delay time of the number of outputs that are switched simultaneously.
- An increase of the delay time of 150 ps to 200 ps for each additional output that is simultaneously switched can be expected.
- With an octal bus driver, such as an SN74xx240, the delay time is increased by 1 ns to 1.4 ns when all eight outputs switch simultaneously.



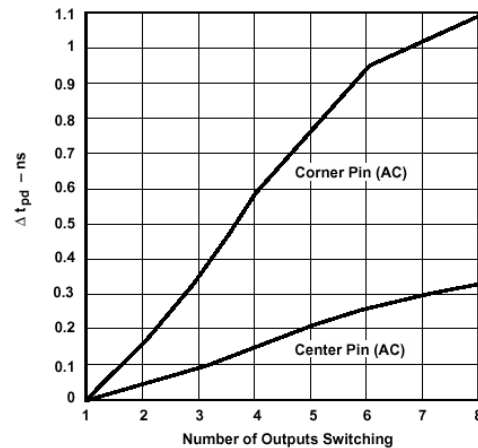


Figure 7.13 Propagation Delay increase with multiple output switching

## 7.12 Propagation Delay Times With Large Capacitive Loads

- In digital logic device data sheets, propagation delay times are specified with a capacitive load of 50 pF (15 pF on older logic families). This value represents the capacitive load of the test circuit on the output of the device being tested.
- This value is also the capacitive load when the output drives five inputs of other circuits and this assumes that the length of the connecting lines is only a few centimeters, as is typically the case on printed circuit boards (PCBs).
- With such short lines, the first assumption is that the line itself behaves like a capacitor, which additionally loads the output and influences the propagation delay time accordingly.
- However, with long lines, this assumption leads to errors, because the signal delay is actually determined by the propagation speed of the electrical wavefront along the line.
- In fact, the propagation delay time of the device is determined by the loading of the output, that is, by the characteristic impedance of the line to which it is connected, not by its length or capacitance.
- When driving a line terminated at its end with a resistance of 100Ω, and lengths of 0 (resistor connected directly to the output), 1 m, and 11 m (see below), an SN74LS00 device has the output waveforms shown in Figure 7.14.

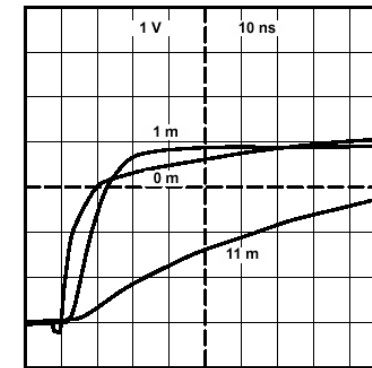
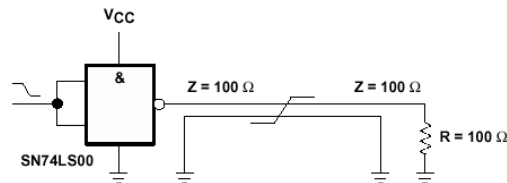


Figure 7.14 Signal delay due to long line

- The length of the line and the resulting signal propagation time (5 ns/m) influence the delay time of the system. The propagation time of the wave along an 11-m transmission line is 55 ns. Add the propagation delay time of the SN74LS00 of about 10 ns for a total delay of 65 ns.

## 7.13 Bus Contention

- If several bus drivers with 3-state outputs are connected to a single bus, it often cannot be ensured that during the time when switching from one bus driver to another, both are not simultaneously active for a short time.
- For this short time, a short circuit of the outputs exists, resulting in an overload of the circuit. This situation is known as *bus contention*.

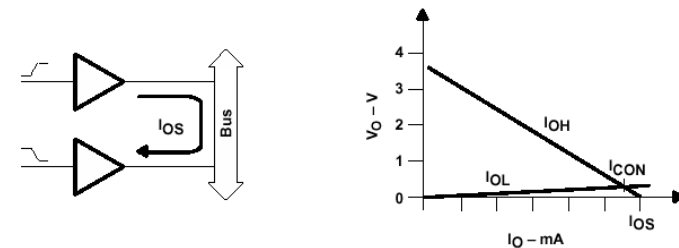


Figure 7.15 Bus Contention

- In cases in which bus contention can occur, which is in most bus applications, power dissipation is of particular importance.
- There is usually nothing that can be done about dynamic conditions (the frequency of operation) without adversely affecting the performance of the system. Also, the

overlap of operating states cannot always be prevented if worst-case conditions are taken into account.

- However, bus contention of a duration of a few, or of a few tens of nanoseconds should not be a problem.
- The choice of the most suitable components allows control of the quiescent power dissipation. With fast bipolar logic families (SN74S, SN74AS, and SN74F), the permissible total power dissipation might be exceeded because of their high quiescent power dissipation.
- Better, in this respect, are the SN74LS and SN74ALS series because, with their lower quiescent-current requirements, bus contention does not result in over-dissipation in most cases.
- Even better are devices from the BiCMOS series and all CMOS devices, although in CMOS parts, a part of the advantage of their low quiescent current is lost by their higher dynamic power dissipation.
- One critical application area is bus contention during the power-on phase of a system. This bus contention occurs because, during the power-on phase (system reset), the supervising circuit does not provide defined control signals even though the rest of the system may already be functional. Therefore, there is a high probability that various bus drivers might be accidentally activated at the same time.
- This, again, results in bus contention that can last several 100 ms (duration of the power-on phase or reset time). Because the thermal time constant of a bus interface device is about 1 ms to 5 ms, after this time expect the final temperature in the device to be determined by momentary power dissipation.
- If a total power dissipation of 5 W in an 8-bit device during this bus contention is assumed, a theoretical 500°C overtemperature of the chip must be considered.
- Even if no defect is detected after such bus contention, a dramatic degradation of the device is likely, which leads to a final destruction of the component some time later.
- Preventing bus contention is not easy because no defined supply voltage can be expected during the power-on phase. Therefore, no defined operation of the logic circuits can be expected.
- One method is to connect a pullup resistor between the enable inputs (assuming low active) of the bus-interface circuits and the positive supply rail. This may ensure a high level as long as the preceding control logic does not provide a defined logic level, but is not helpful if the control logic delivers a wrong logic level.
- A reliable solution is to disable all bus-interface circuits in question during the critical time with additional control logic

### 7.14 Pullup or Pulldown Resistors

- When buses are disabled for more than the maximum allowable time, other ways should be used to prevent components being damaged or overheated.
- A pullup or a pulldown resistor to  $V_{CC}$  or GND, respectively, should be used to keep the bus in a defined state. The size of the resistor plays an important role and, if its resistance is not chosen properly, a problem occurs.
- Usually, a 1-k $\Omega$  to 10-k $\Omega$  resistor is recommended. The maximum input transition time must not be violated selecting pullup or pulldown resistors (see Figure 7.16). Otherwise, components may oscillate, or device reliability may be affected.

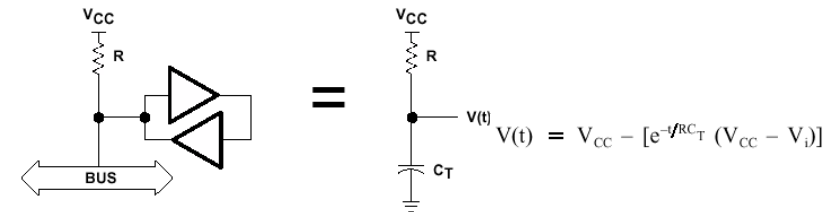


Figure 7.16 Effect of pull-up resistor

- This pullup resistor method is recommended for ac-powered systems; however, it is not recommended for battery-operated equipment because power consumption is critical.
- Instead, use the bus-hold feature that is discussed in the next section. The overall advantage of using pullup resistors is that they ensure defined levels when the bus is floating and help eliminate some of the line reflections, because resistors also can act as bus terminations.

### 7.15 Bus Hold Circuit

- Bus-hold circuits are used in selected TI families to help solve the floating-input problem and eliminate the need for pullup and pulldown resistors.
- Bus-hold circuits consist of two back-to-back inverters with the output fed back to the input through a resistor (see Figure 7.17).
- To understand how the bus-hold circuit operates, assume that an active driver has switched the line to a high level. This results in no current flowing through the feedback circuit.
- Now, the driver goes to the high-impedance state and the bus-hold circuit holds the high level through the feedback resistor.
- The current requirement of the bus-hold circuit is determined only by the leakage current of the circuit.
- The same condition applies when the bus is in the low state and then goes inactive.



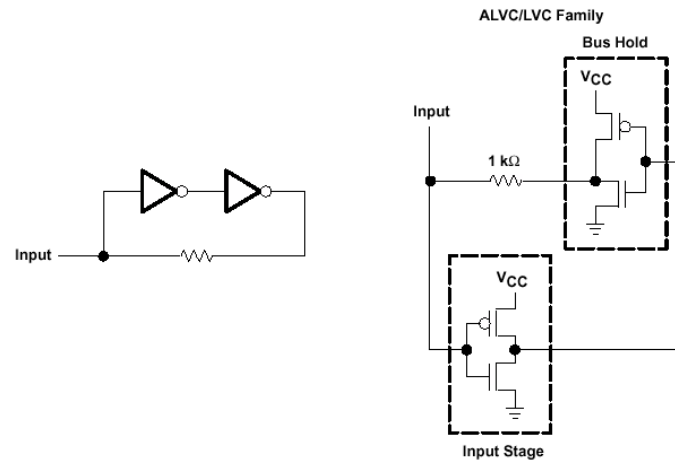


Figure 7.17 Bus Hold Circuit

### 7.16 The Problem of Ground Bounce

- Ground Bounce is a voltage oscillation between the ground pin on a component package and the ground reference level on the component die.
- Ground Bounce is one of the primary causes of false switching in high speed components and is a major cause of poor signal quality.
- Essentially it is caused by a current surge passing through the lead inductance of the package.
- The effect is most pronounced when all outputs switch simultaneously, (hence the alternate name, Simultaneous Switching Noise).
- While the inductance is the combined effect of the package lead, the package lead frame, the bond wire and the inductance in the die pad, most of the inductance is caused by the bond wire.
- Figure 7.18 shows a typical waveform for a high speed CMOS component with TTL level outputs. Shown directly above the signal waveform on the same time scale is the voltage seen on the ground of the die relative to the external ground.
- Directly above that is the voltage seen on the die Vcc relative to the external Vcc on the board.

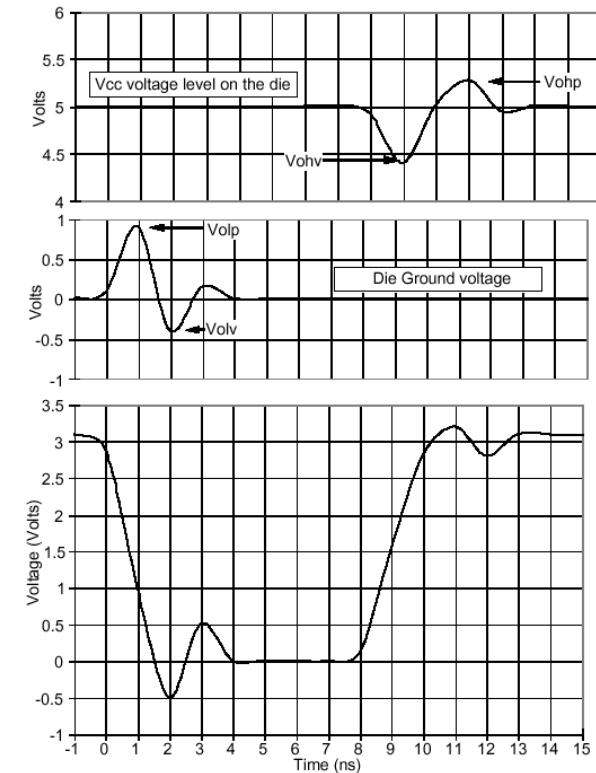


Figure 7.18 Ground and Vcc Bounce

- Ground Bounce levels are typically more pronounced than Vcc Bounce levels because of the HIGH to LOW transition is trying to quickly bring a HIGH signal down to a narrow window of <400mV for a logic LOW. A low output impedance is required to complete the transition quickly.
- In the LOW to HIGH, the only requirement is that the output be above 2.4V. 5V is available as a driving voltage. A much lower pull up impedance is required to make the transition quickly.
- Ground Bounce effect can be reduced by:
  1. Using more Vcc/Gnd pins (offered by manufacturers)
  2. Avoid pullup/pulldown resistors (i.e. use bus hold circuit)
  3. Possibly use series damping resistors (see later)
  4. Control output slew rate
  5. Extra care in holding clock signals to solid voltage level (Vh) and fast clock transitions

### 7.17 Series Output Damping Resistors

- The purpose of integrating output-damping resistors in line buffers and drivers is to suppress signal undershoots and overshoots on the transmission line through what is usually referred to as line-impedance matching (see Figure 1).
- The effective output impedance of the line driver ( $Z_O$ ) is matched with the line impedance ( $Z_L$ ). Thus, no signal reflection occurs at the line start.
- The input impedance of the receiving device ( $Z_I$ ) is assumed to be several orders of magnitude higher than the line impedance. This is valid for CMOS and BiCMOS devices. In this case, the reflection coefficient at point B is approximately 1, such that almost all of the wave energy is reflected at the end of the line.

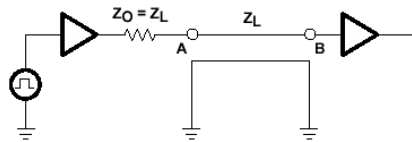


Figure 7.19 Transmission line effect

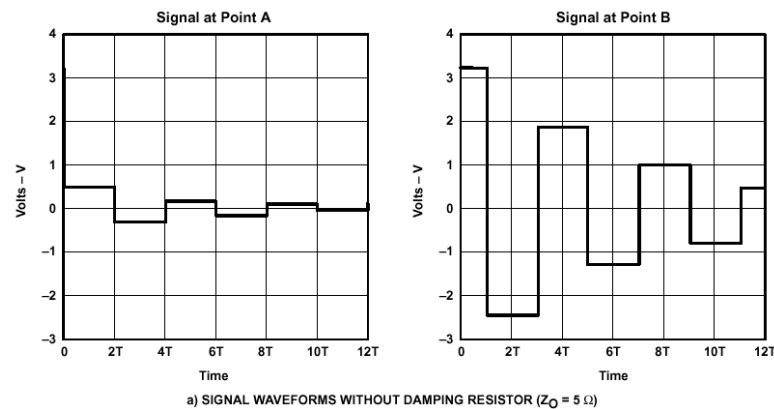


Figure 7.20 Transmission line effect

- To reduce this ringing effect, output damping resistors can be added:

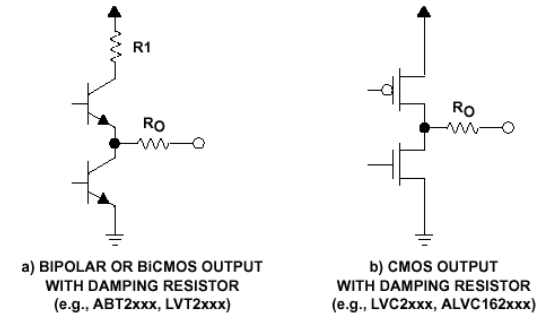


Figure 7.21 Output damping resistors

- The effect of this resistor (typically  $25\text{--}33\Omega$ ) is to eliminate most of the ringing effect:

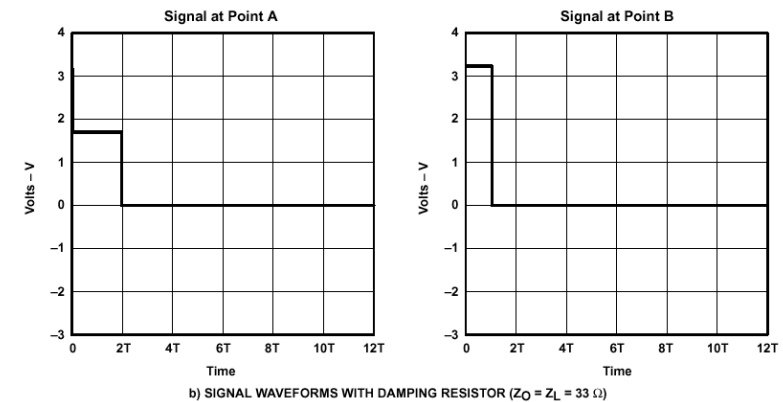
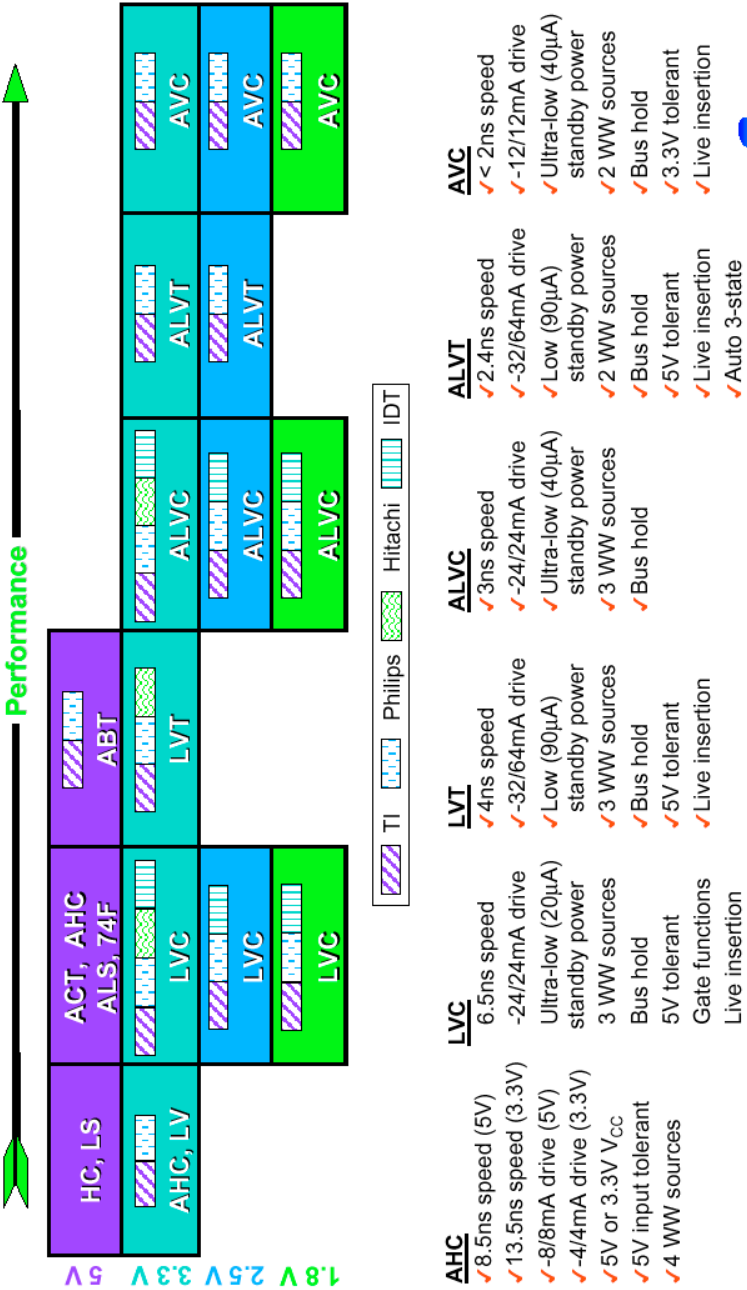


Figure 7.22 Effect of damping resistors

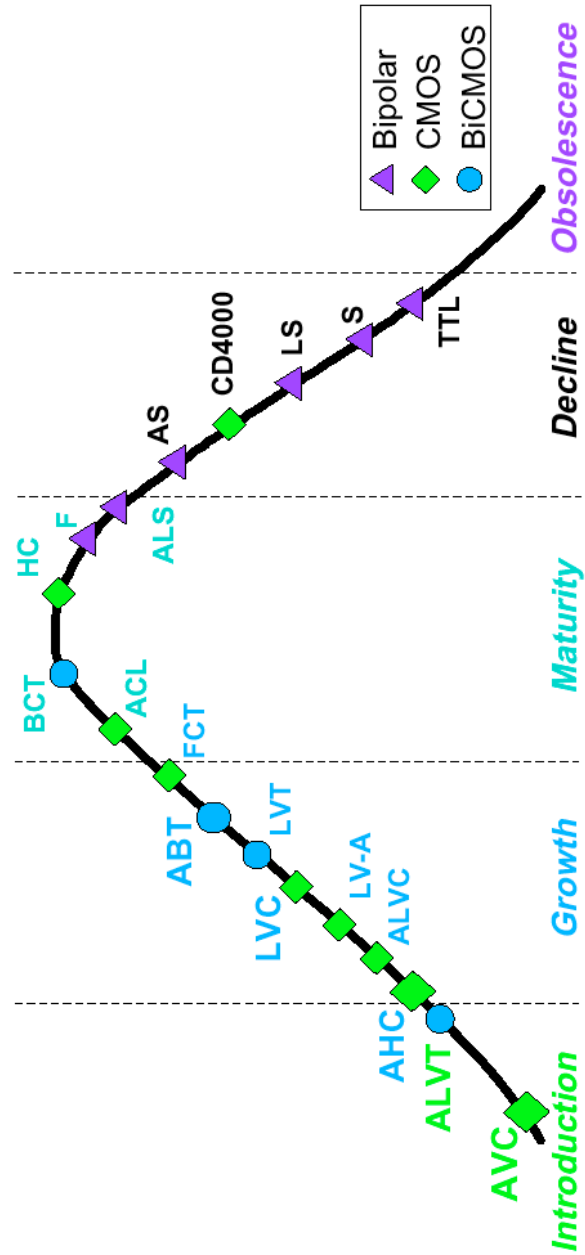
7.18 Summary of the various logic families (from TI)

|      | $\Delta t/\Delta t^\dagger$<br>(ns/V) | Drive<br>$I_{OL}/I_{OH}$<br>(mA) | $I_{CCD}$<br>(mA/<br>MHz<br>Bit) | TYP<br>$V_{OLP}/V_{OHV}$<br>(V) | TYP<br>Output<br>Rise/<br>Fall<br>(ns) | TYP<br>$I_{PD}$<br>vs<br>Cap.<br>(ns/<br>pF) | TYP<br>$I_{PD}$<br>vs<br>SS<br>(ns/<br>Nsw) | MAX<br>$I_{PD}$<br>(ns) | TYP<br>Skew<br>(ns) | MAX<br>$I_{CC}$<br>CMOS<br>(mA) | MAX<br>$I_{CCH}/I_{CCL}/I_{CCZ}$<br>BICMOS<br>(mA) | 5-V<br>Tol.    | PU3S           | Bus<br>Hold    | 25- $\Omega$<br>Output<br>Series<br>Resistor | TYP<br>Control/<br>I/O<br>Cap.<br>(pF) |
|------|---------------------------------------|----------------------------------|----------------------------------|---------------------------------|----------------------------------------|----------------------------------------------|---------------------------------------------|-------------------------|---------------------|---------------------------------|----------------------------------------------------|----------------|----------------|----------------|----------------------------------------------|----------------------------------------|
| LV   | 100                                   | 8/-8 <sup>†</sup>                | 0.22                             | 0.70/<br>2.4                    | 1.81/<br>2.44                          | 0.028                                        | 0.11                                        | 16                      | 0.74                | 0.02                            |                                                    |                |                |                |                                              | 2.5/<br>2.5/7                          |
| LVC  | 10                                    | 24/-24                           | 0.23                             | 0.57/<br>2.4                    | 0.54/<br>0.6                           | 0.014                                        | 0.04                                        | 5.2                     | 0.27                | 0.02                            |                                                    | √ <sup>†</sup> |                | √ <sup>†</sup> | √ <sup>†</sup>                               | 3.3/<br>5.5/9                          |
| LVT  | 10                                    | 64/-32                           | 0.12                             | 0.38/<br>2.47                   | 0.47/<br>0.49                          | 0.01                                         | 0.04                                        | 4.1                     | 0.48                |                                 | 0.12/<br>5/<br>0.12 <sup>†</sup>                   | √              |                | √              |                                              | 3.5/<br>4/10                           |
| LVTZ | 10                                    | 64/-32                           | 0.12                             | 0.38/<br>2.47                   | 0.47/<br>0.49                          | 0.01                                         | 0.04                                        | 4.1                     | 0.48                |                                 | 0.12/<br>5/<br>0.12 <sup>†</sup>                   | √              | √              | √              |                                              | 3.5/<br>4/8                            |
| LVT2 | 10                                    | 12/-12                           | 0.12                             | 0.34/<br>2.6                    | 0.5/<br>0.5                            | 0.018                                        | 0.05                                        | 4.9                     | 0.35                |                                 | 0.12/<br>5/<br>0.12 <sup>†</sup>                   | √              |                | √              | √                                            | 3.5/<br>4/10                           |
| ALVC | 10                                    | 24/-24                           | 0.27                             | 0.70/<br>2.38                   | 0.33/<br>0.29                          | 0.014                                        | 0.05                                        | 3.4                     | 0.45                | 0.04                            |                                                    |                |                | √              | √ <sup>†</sup>                               | 3.5/<br>6/7.5                          |
| ABT  | 10                                    | 64/-32                           | 0.49                             | 0.54/<br>3.3                    | 0.52/<br>0.42                          | 0.013                                        | 0.04                                        | 4.2                     | 0.25                |                                 | 2/<br>30/<br>2 <sup>†</sup>                        |                | √ <sup>†</sup> | √ <sup>†</sup> |                                              | 3/<br>3/6                              |
| ABT2 | 10                                    | 12/-12 <sup>†</sup>              | 0.49                             | 0.40/<br>3.3                    | 0.62/<br>0.55                          | 0.019                                        | 0.04                                        | 4.2                     | 0.18                |                                 | 2/<br>30/<br>2 <sup>†</sup>                        |                | √ <sup>†</sup> | √ <sup>†</sup> | √                                            | 3/<br>3/6                              |
| ABTE | 10                                    | 90/-60                           | 0.42                             | 0.70/<br>3.1                    | 0.75/<br>0.47                          | 0.024                                        | 0.04                                        | 5.2                     | 0.11                |                                 | 36/<br>48/<br>32                                   |                |                | √              |                                              | 2.5/<br>2.5/4.5                        |
| GTL  | 10                                    | 40 <sup>§</sup>                  | 0.30                             | 0.40/<br>1.1                    | 0.64/<br>0.56                          | 0.0054                                       | 0.04                                        | 4                       | 0.17                |                                 | 120                                                | √              |                | √              |                                              | 3.5/<br>3.5/4                          |
| FB   | 5                                     | 100 <sup>§</sup>                 | 0.20                             | 1.3/<br>1.9                     | 0.43/<br>0.37                          | 0.014                                        | 0.04                                        | 5.6                     | 0.30                |                                 | 60 <sup>†</sup>                                    |                | √              |                |                                              | 5/<br>5/4                              |
| CBT  | -                                     | 0/0                              | 0                                | 0.14/<br>2.89                   | 0.64/<br>0.56                          | 0.013                                        | 0.02                                        | 0.25                    | 0.23                | 0.003 <sup>†</sup>              |                                                    |                |                |                |                                              | 3/<br>6/6                              |

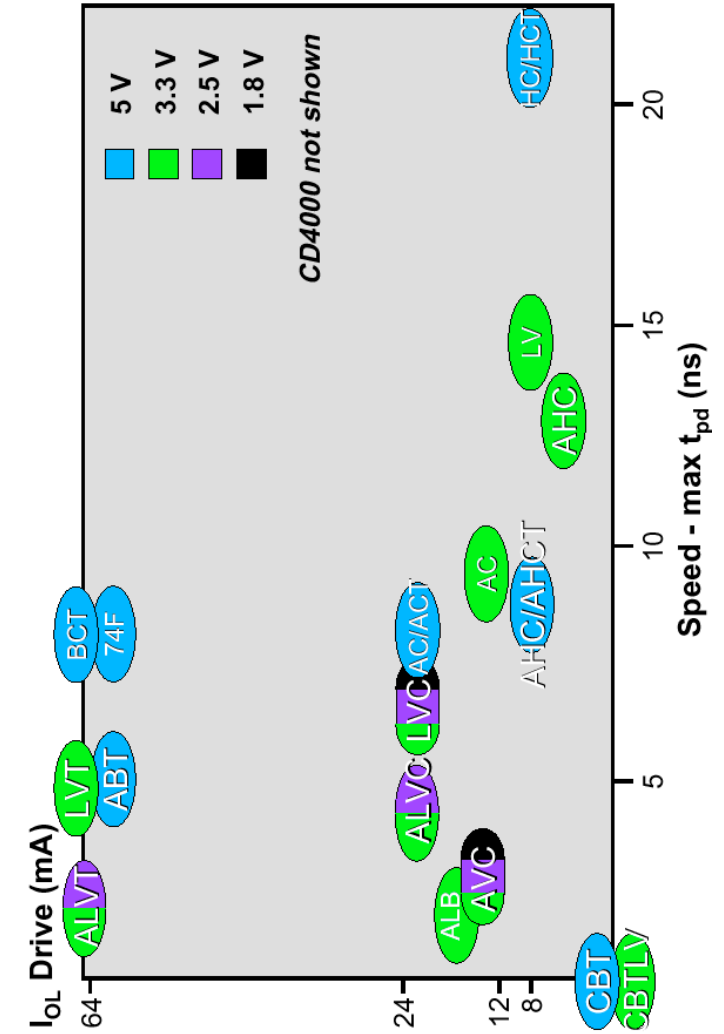
Complete Low-Voltage Market Coverage and Standardization



# Product Life Cycle



# Family Performance Positioning



## References:

1. "Understanding Advanced Bus-Interface Products", TI AppNote SCAA029, May 1996
2. "Designing With Logic", TI AppNote SDYA009C, June 1997
3. "Bus-Interface Devices With Output-Damping Resistors" TI AppNote SCBA012A, August 1997
4. "The Myth of Ground Bounce Measurements and Comparisons", IDT AppNote AN-147, 1996.